

Lab 01: Using Equipment

Lab Training & Equipment Demonstrations

Learning Objectives

Upon successful completion of this lab, students should know how (to)...

- Safely interact with benchtop equipment, 3D printers, power tools, and other lab equipment
- Use an oscilloscope, function generator, power supply, and multimeter for testing;
- Interface with test equipment through a computer to collect data.
- Debug hardware problems through test-driven troubleshooting; and

Overview

In this activity, you will be trained in using the various lab equipment and safety procedures of the lab, learn to use the standard equipment used for analog and digital computer development (i.e. the benchtop lab equipment), demonstrate use of the benchtop lab equipment, and complete a report which will require you to use some of the equipment.

Safety and lab procedure trainings **must** be complete *before* working with **any** equipment. These trainings prepare you to safely utilize the lab equipment for manufacturing, assembly, and debugging within the lab. See lab manager for more information.

You will demonstrate your understanding of the lab equipment and procedures by completing the following:

- *Guidelines Quiz*: Some questions regarding the lab's safety and operating procedures.
- *Equipment Demo*: A couple short exercises demonstrated to a TA (see below). You are able to practice on your own beforehand. The demonstration, at it's minimum, is described below and usually takes no longer than 5 minutes.
- *Report*: You will prepare a lab report that evaluates your understanding of the benchtop equipment as well as your usage of it to create a gain Bode plot.

Specification

After learning to use the equipment, you will need to

- 1) complete the guidelines quiz;
- 2) demonstrate your ability to use equipment during lab hours before the due date; and,
- 3) complete exercises in the document and submit a report.

Note: If many people show up on the same day to demonstrate, you may have to wait – and may not be able to complete it on time as a result! There **will not be special consideration given** to demonstrations delayed due to capacity issues in the days of and just before the due date.

This demonstration will be completed in *at least* two steps:

1) Wave Generation & Display

Generate a 1kHz, ± 1 V peak square wave with a +100mV offset using the function generator. Display the resulting waveform on the oscilloscope by adjusting position, scale, and trigger settings until displayed waveform is not drifting and at least two full periods are on-screen. Trigger on the rising edge of the waveform. The full positive and negative peaks of the waveform should be visible.

2) Voltage Test

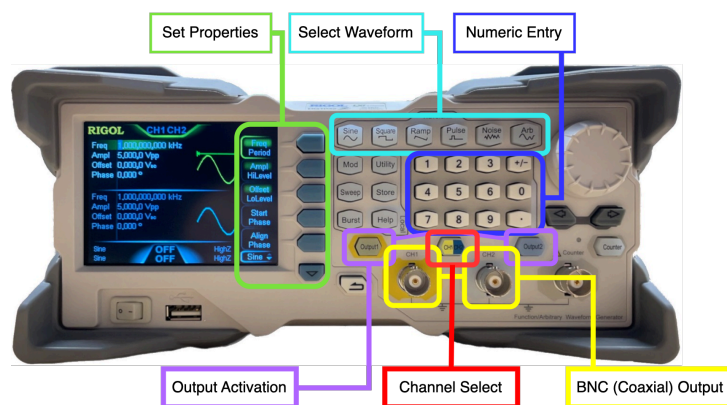
Using the +5V max channel of the power supply, determine the voltage needed to supply 100mA of current to a 10Ω resistor. Verify by placing the digital multimeter (in current meter mode) in series with the resistor and power supply set to the correct voltage. Set the power supply current limit to 50mA and turn up the voltage. Show that current through the resistor does not exceed 50mA.

Equipment

This section describes the equipment in the lab, what it does, and how to use it.

Function Generator

The **function generator** is typically used to generate periodic (AC) waveforms and arbitrary waveforms (noise). It can be used to generate input signals for testing frequency-dependent circuits (e.g., filters or amplifiers) and clock signals for digital circuits. [Manual] [Programming]



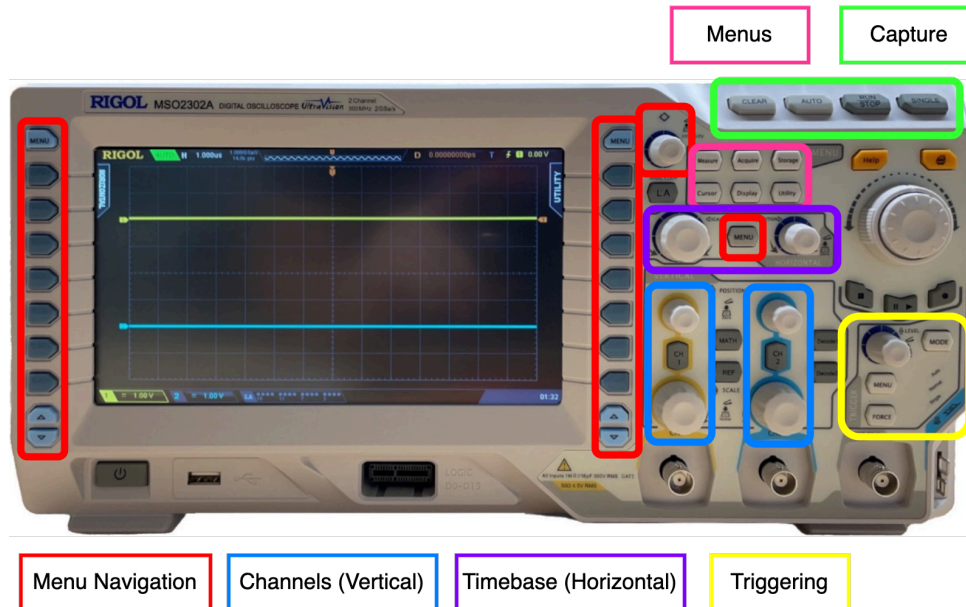
BNC to Alligator Clip Cable

Users can select a **waveform** (e.g., sine, square, and ramp) and set **amplitude**, **frequency**, **period**, **phase**, and **offset**. This model can set up two outputs; we can sconnect a coaxial cable for each.

The cable shown on the right is the typical way of connecting the function generator outputs to a test circuit. Attach the BNC connector (coaxial cable connector) to the instrument, the black alligator clip to the circuit ground, and the red alligator clip to the circuit input.

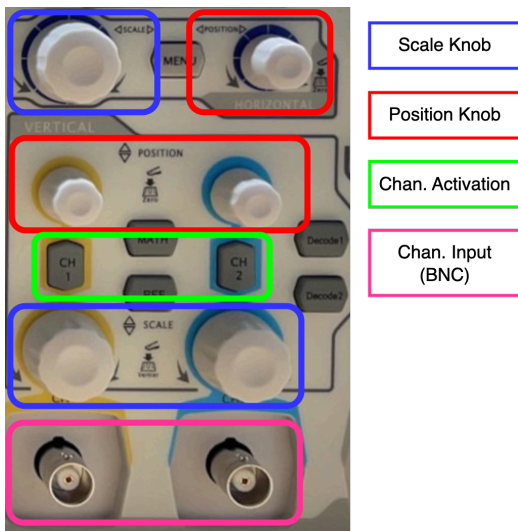
Digital Oscilloscope

The **oscilloscope** can measure all kinds of waveforms (e.g. from an oscillator, filter, or digital protocol such as I2C or SPI). Features include triggering (various types), scaling & positioning (time & voltage), and coupling (for measuring AC signals accurately on top of a DC offset). [Manual] [Programming]



Channels & Viewing Captures

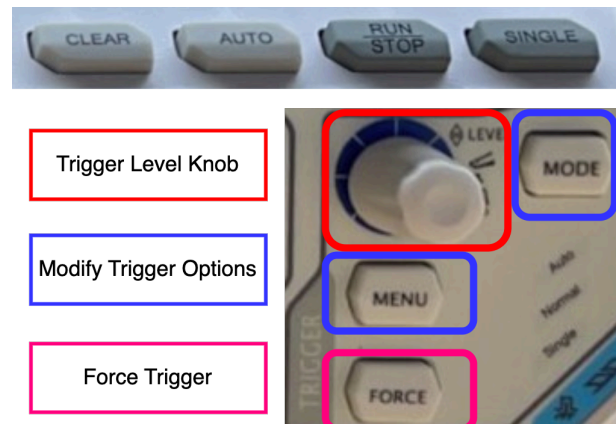
The channels share the **timebase** (*horizontal axis*) scale/position, but each have their own **signal range** (*vertical axis*) scale/position.



A *position* defines an offset, while a *scale* defines a linear scaling (combines to form a linear transformation).

Capture Controls & Modes

There are two main ways to capture waveforms: via a **SINGLE** capture or via a **RUN-STOP** sequence (continuous mode).



In *single* capture mode, unsurprisingly, a single capture (recording) is taken when the trigger is activated, and the triggering is stopped and the capture displayed.

In *continuous* capture mode, data is continuously recorded until the STOP key is pressed, at which

Using Timebase & Signal Range

While all four timebase/signal range – position/scale knobs can be used to *digitally transform* a captured signal, three of the knobs actually affect signal acquisition (i.e. at capture-time)!

Only the **timebase position** is purely a *display operation*: the trigger event defines $t = 0$, the scope stores samples on either side of it, and the knob merely shifts the displayed time window of the existing capture. The other three reach into the analog front end before the ADC (analog-to-digital converter):

- **Timebase scale** sets the sample rate and record length, fixing the temporal resolution and the total capture window.
- **Signal range position** drives an offset DAC summed with the input. Push the trace far enough off-screen and the signal leaves the converter's input window and is captured clipped – the recorded data is wrong, not merely out of view.
- **Signal range scale** sets the attenuator and amplifier gain, determining how much of the ADC's range the signal occupies, and hence the effective vertical resolution.

Important: A practical consequence is that position and scale are not cosmetic – a signal must be centered and made to fill a good fraction of the screen *before* capture; no amount of post-hoc digital panning or zooming recovers detail that was clipped or never sampled.

point triggering is stopped and the capture is displayed.

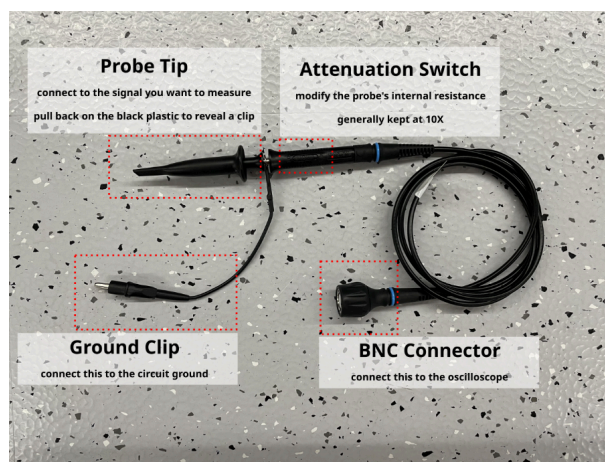
Additionally, the **CLEAR** key can be used to clear the recorded data.

Math & Additional Options

The oscilloscope also has built-in functions that can be performed on recorded data via the **math** button – e.g., Fast Fourier Transforms (FFTs). There are also additional options / menus for adjusting waveform capture.



Oscilloscope Probe

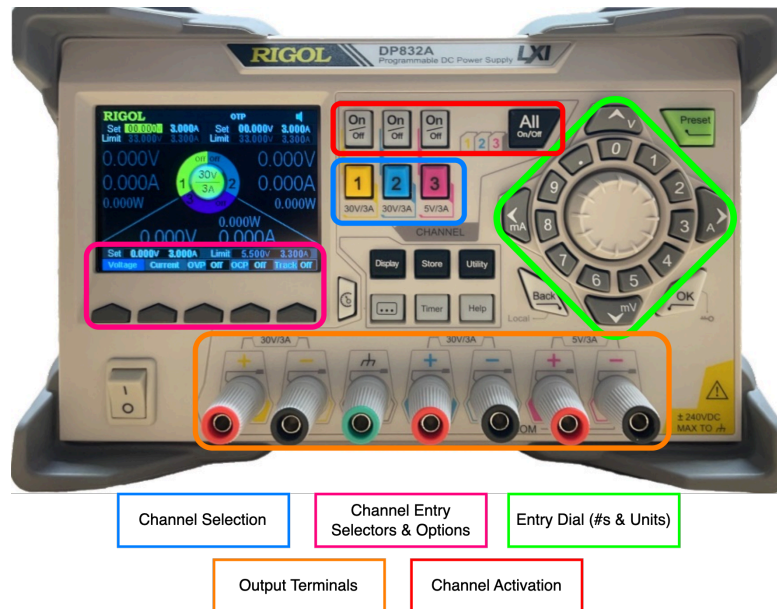


The oscilloscope **probe** is the proper connector to use to use the scope to measure circuit signals. The **probe tip** is used to *probe* some measurement point on a circuit, and the **ground clip** is used to establish a reference voltage to compare against (i.e. ground / 0V).

The **attenuation switch** modifies the internal resistance of the probe, and in general is kept in the 10x position (meaning the voltage fed into the instrument is reduced by a factor of 10 from the true value by virtue of a voltage divider). This is used to increase the impedance of the probe to minimize its effect on the operation of the circuit. The **channel ratio** must be set to the attenuation factor of the probe you are using to display correctly scaled voltages. Find this in the channel's corresponding menu.

Power Supply

The **power supply** provides DC power (voltage or current) to a circuit. It allows the user to set the voltage / current output and limit the voltage output/current to supply a circuit (e.g., a 5V supply for a chip, or an input voltage to test a voltage regulator). [Manual] [Programming]



Our power supplies have 3 channels – one with a max of 5V, and two at 30V – with all up to 3A. Each channel has a selection button (to make it the current channel for entry), terminal outputs (for voltage / current out), and an activation button (on / off). There is also a button to turn all channels on / off. There is also a “true ground” (chassis/plug ground), but this should only be used when floating / relative voltage is not sufficient for the application.

To set a constant voltage or current for the selected channel, choose “Voltage” or “Current” from the on-screen menu. Once selected, use the numeric pad to enter the value, and select the units using the “units” buttons. You can return to the previous digit using the “Back” button or reset the units using the “OK” button.

Digital Multimeter

While the multimeter is generally only useful for checking the steady-state values, our model does have the ability to measure operating point of a circuit (power supply voltage/current) or resistance/continuity between two points (e.g., verifying output of a voltage regulator, checking for shorts between different traces on a printed circuit board before operation, or checking continuity on traces that are supposed to be connected.)



Figure 9: Labelled above are the primary arrangements of connectors for measuring various values.

The images to the right show the typical cables used with the multimeter and power supply. The left image shows simple **banana-to-alligator** leads. Use them to connect power and ground from the power supply to the test circuit. The image on the right shows the **multimeter test probes**. Use them to accurately probe a test point (and ground point) on the circuit being measured.



Remote Control

Electrical test equipment has historically included a means of control through an external interface. One of the first implementations of this technique was by HP in the 1960s – today known as the GPIB (General Purpose Interface Bus). Over time, engineers designed what is now known as the Standard Commands for Programmable Instrumentation (SCPI) protocol. The equipment in the lab supports this protocol through more modern electrical interfaces such as USB, TCP/IP, and RS-232 (serial).

```
user@host ~/labs/eq_demo> echo ":SYST:TIME?" | nc -w1 10.245.26.231 5555
15,51,49
```

Listing 1: Example of sending an SCPI command using netcat from a bash shell.

Each set of benchtop equipment are on their own private LAN network. We can consult the product manual to learn how to find the IP address of our instruments (in 169.254.0.0/16). For example, the MSO2302A Oscilloscope's IP address can be through the **Utility** → **LAN set** button sequence.

To be able to connect to them you must connect your computer to the network's packet switch via ethernet. If your computer does not have an ethernet port, use must you an ethernet to USB-C adapter (USB-C to USB-A adapters are available, upon request). Ethernet adapters may be found in the lab's stock; their cabinet is clearly marked, and it is your responsibility to find them.

Note: You *may* need to change the ethernet link's configuration in your OS so that DHCP is disabled. We use local-link IPV4 autoconfiguration since our private networks don't have DHCP servers (see here for more info).

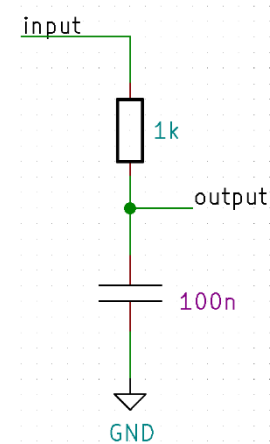
Report

Students should complete these exercise prompts in their reports.

- 1) **Warm-up:** Explain the terms **volts per div** and **time per div** as applied to an oscilloscope. Where specifically are they indicated on the scope's display?
- 2) **Warm-up:** What's the **green** terminal on the power supply? What can it be used for and how, physically, could it be used?
- 3) **Technical:** Explain, in your own words, how the *scale* settings for voltage & time affect an oscilloscope's capturing of a waveform. (A good response should identify, describe, and explain the inner workings of an oscilloscope.)
- 4) **Scenario:** You are designing a class-D audio amplifier that needs a bipolar, +/-12V DC power supply for the MOSFETs, as well as a 5V DC supply to power to logic circuitry. Make a labelled diagram showing how this could be done with power supplies available in our lab (**note:** all physical connections should be shown!).
- 5) **Scenario:** Rob received a synth PCB he designed by mail. He assembled the board and baked it in the reflow oven. He plugged it in and... pop! The power supply was reversed. *Oops!!*
 - a. Briefly describe how a programmable power supply and other equipment could detect which component(s) were damaged with minimal risk (hint: faulty components get hot). (Must be practical.)
 - b. With your help, Rob was able to find and replace the broken parts. He can program his microcontroller, but the SPI interfaced DAC isn't working as he thinks it should. Briefly describe a technique he could use to debug the circuit.
- 6) **Scenario:** You are a test engineer for a drone manufacturer. The flight-ops team describes incidents where drones crash after an aggressive flight maneuver. You think the high motor load during maneuvers is causing the microcontroller to brown out. How could you test this hypothesis?

5) **Practice:** Construct this simple, passive low-pass filter on a breadboard.

- Calculate the theoretical cutoff frequency of this filter. Show your calculations.
- Write a Python script to collect data on the frequency response of the filter. Starting below the cutoff frequency and ending above, use SCPI to iterate the function generator through a series of frequencies and acquire the attenuation from the oscilloscope. Capture at least 32 data points. Save the data in a file and use **matplotlib** to create a Bode plot for the circuit with (correct) labelled axes, a title, and a marker of the experimental cutoff frequency. The Bode plot should be in the report.
- Measure and document the resistor and capacitor values using the multimeter and use the values to calculate the actual cutoff frequency. Does this match the frequency from the Bode plot? If not, what might account for discrepancies?



Submission

Your submission will consist of the following elements:

- **Canvas (5 pts):** Completion of lab guidelines quiz in Canvas.
- **In-lab (10 pts):** Demonstration of your test-equipment knowledge.
- **Gradescope (15 pts):** Report in **.pdf format** containing your answers to the exercises above (use `uf-cpe-capstone/report-template-latex`) & Python script as a **.py file**.